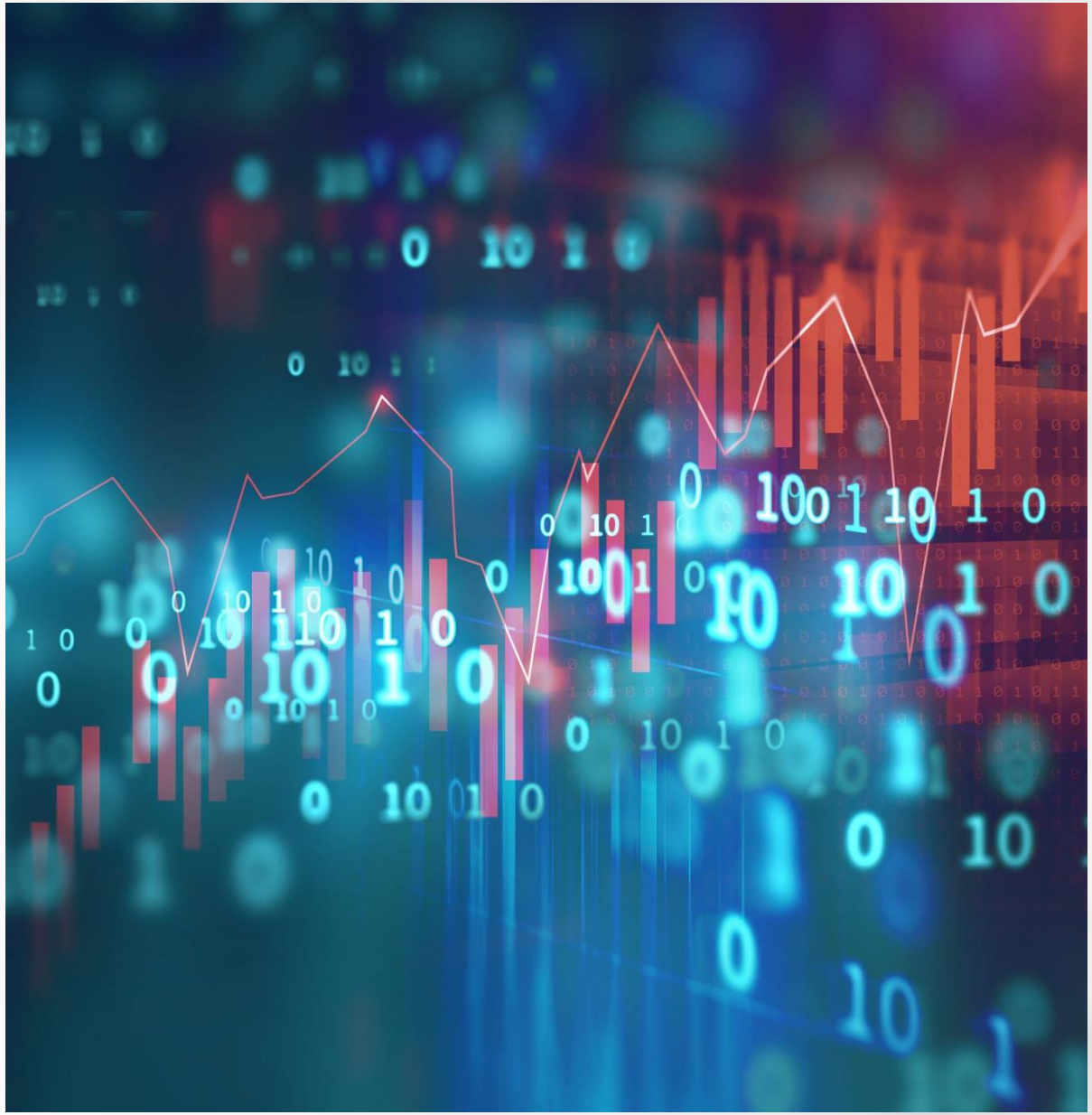


Data Analysis of Developer Survey Data

DENİZ ÖZTÜRK

16 March 2026

OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- The job skills analysis shows that programming languages and technical tools are strongly linked to the number of job postings, indicating that employers prioritize candidates with practical technical skills.
- The Top 10 programming languages visualization shows that a small group of languages dominate the job market.

-Python, JavaScript, and SQL are among the most widely used languages.

-These languages are widely used across multiple data and software development roles.

-Their high demand indicates their importance in the current technology ecosystem.

- The average annual salary analysis indicates that salaries vary significantly depending on the programming language and specialization.
- The dashboards provide an interactive overview of technology trends, helping identify the most in-demand programming languages and skills in the job market.

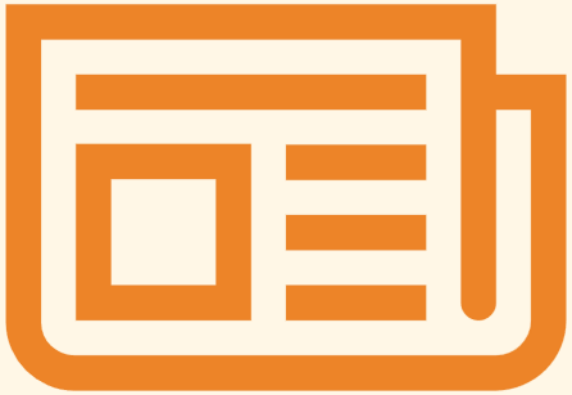
INTRODUCTION

- The purpose of this report is to analyze trends in the technology job market using developer job survey data.
- The target audience includes aspiring data professionals, software developers, students, and individuals interested in understanding trends in the technology job market.
- The insights from this analysis help highlight the most relevant technologies and skills required in the industry.

-Job seekers can use these insights to focus on learning programming languages that are highly demanded in the job market.

-Interactive dashboards based on job survey data provide information on programming languages used, databases, and required job skills. It also offers an overview of employee demographics.

METHODOLOGY



- **Data Collection**
- **Data Cleaning and Preparation**
- **Data Analysis**

-Top 10 Programming Languages: Identified the most frequently requested programming languages in job postings.

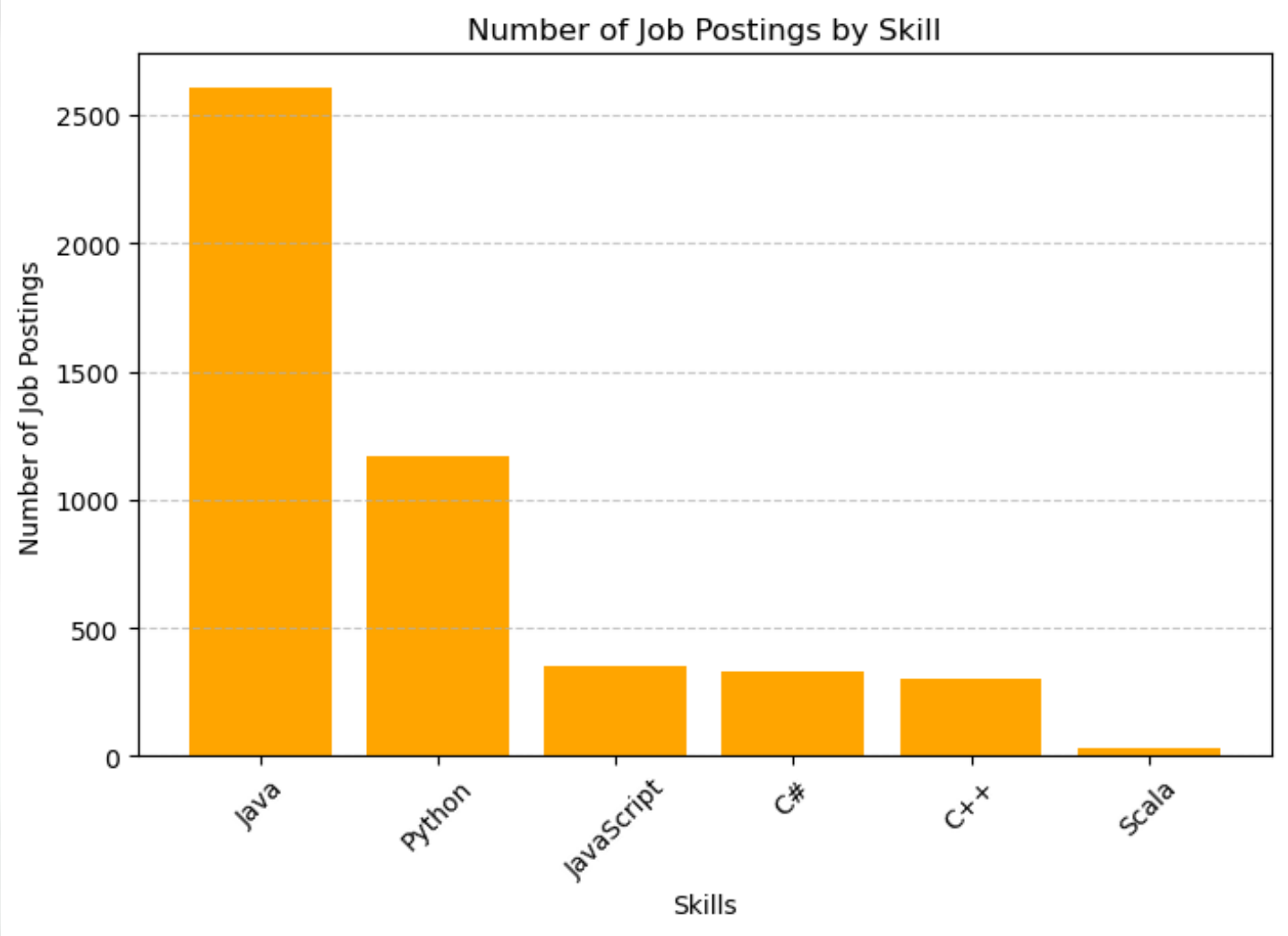
-Average Annual Salary Analysis: Calculated average annual salaries for different programming languages to explore compensation trends.

-Job Skills Analysis: Examined the relationship between technical skills and the number of job postings.

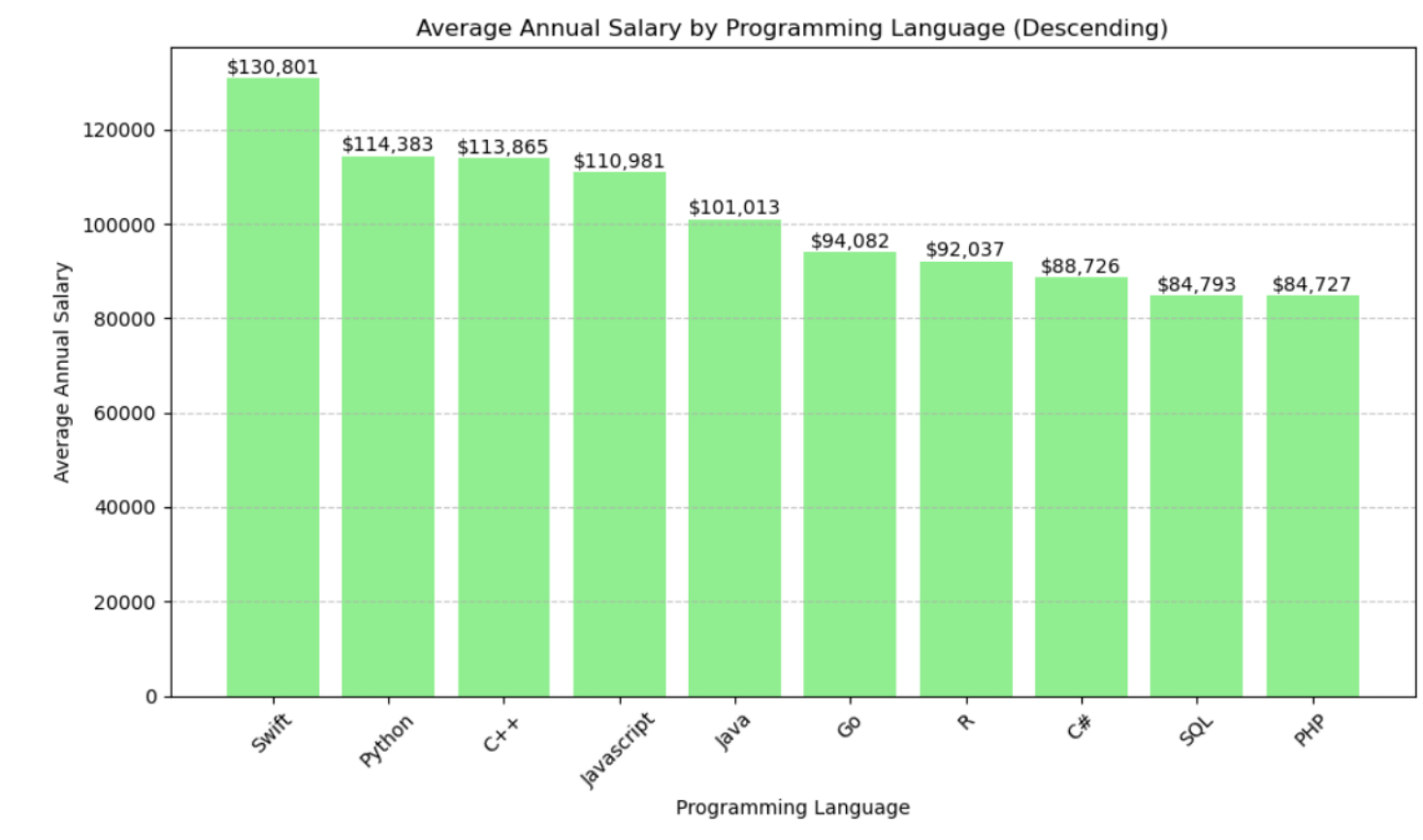
- **Visualization and Dashboards**
- **Interpretation**

Results

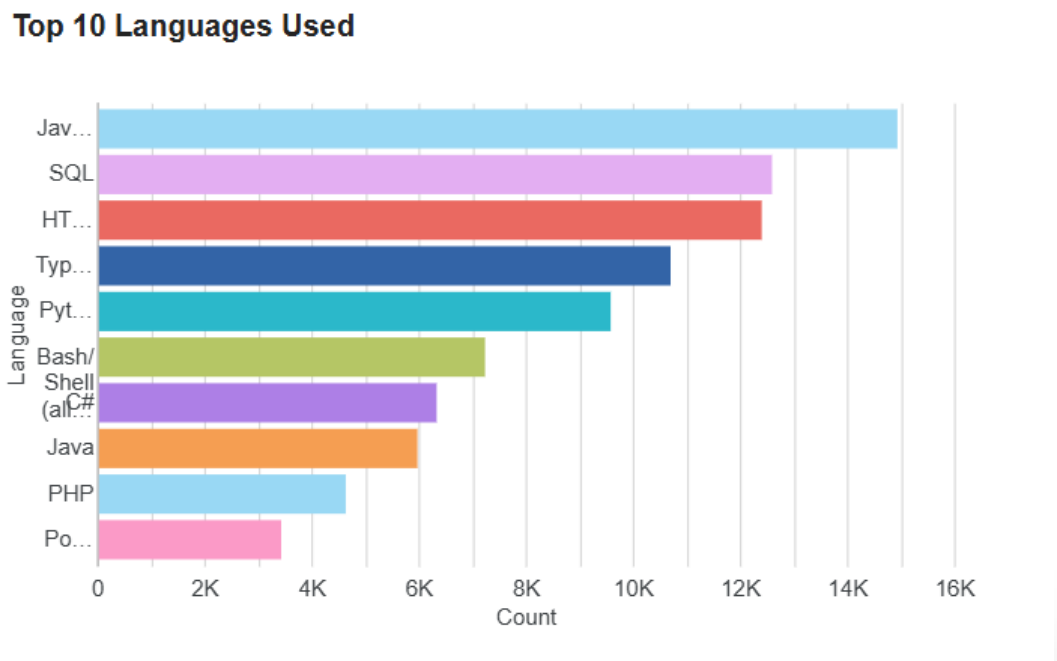
JOB POSTING

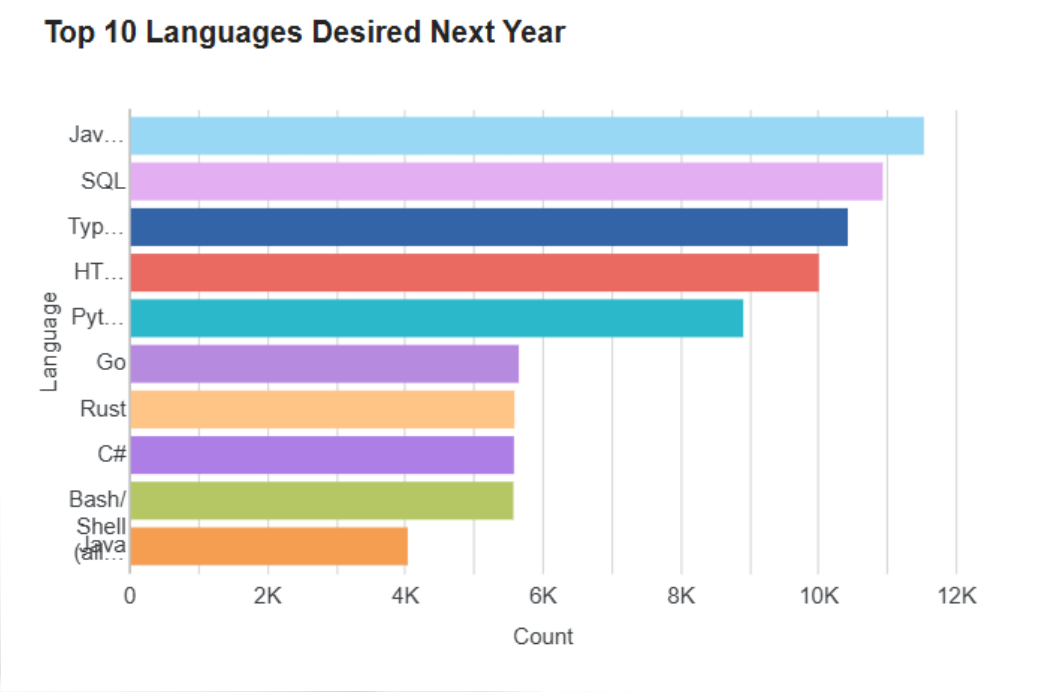

Number of Job Postings by Skill
Number of Job Postings
2500
2000
1500
1000
500
0
Java
Python
JavaScript
C#
C++
Scala
Skills

POPULAR LANGUAGES



PROGRAMMING LANGUAGE TRENDS





PROGRAMMING LANGUAGE TRENDS- FINDINGS & IMPLICATIONS

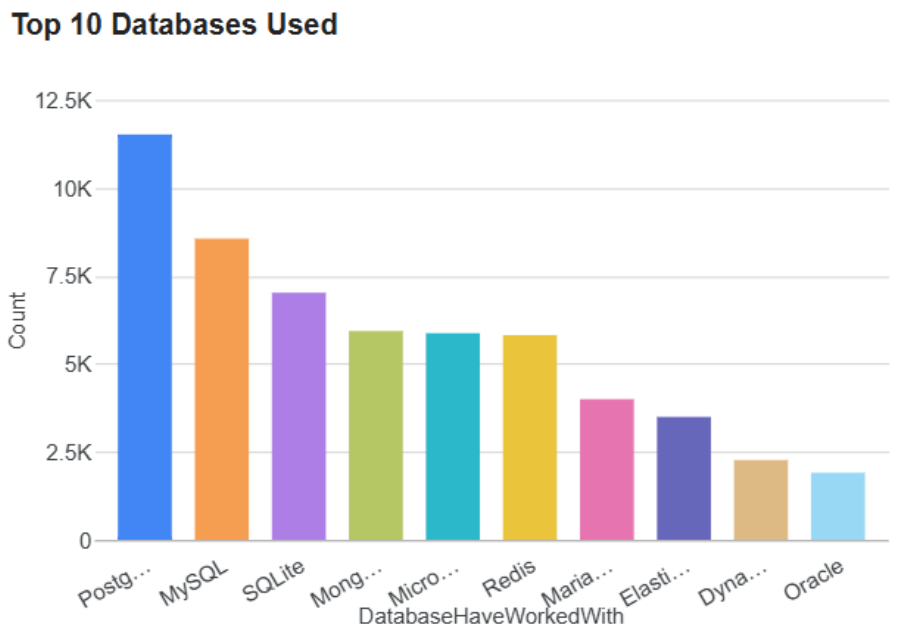
Findings

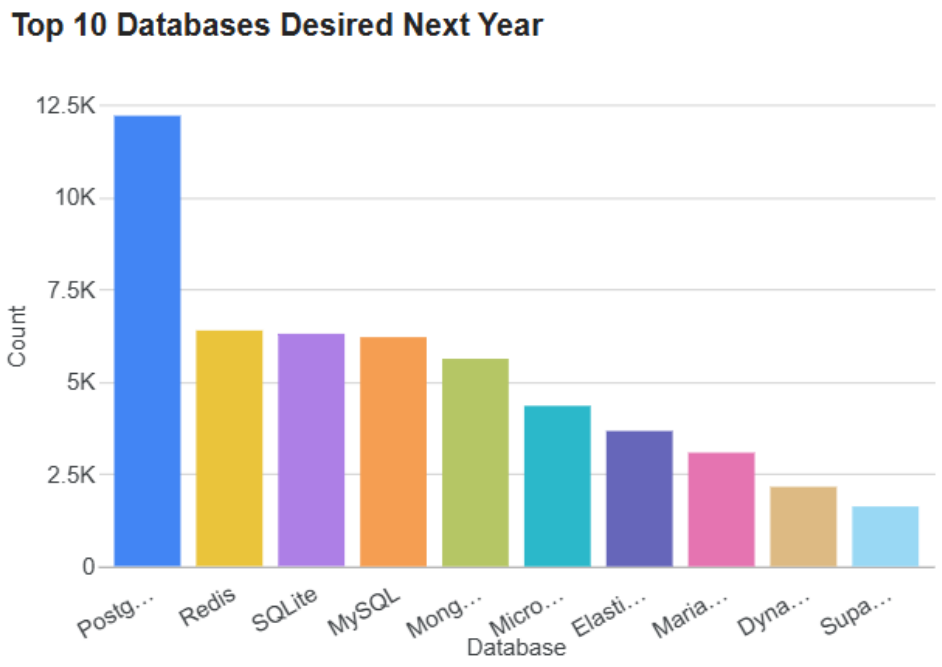
- JavaScript, SQL and HTML/CSS are the most commonly used languages according to current job postings.
- The Top 10 Desired Languages chart shows strong employer demand for Javascript, SQL and TypeScript highlighting slight differences between current usage and employer preferences.
- Overall, the comparison reveals a trend where a few languages are dominant in use, while employer demand leans towards popular languages.

Implications

- These languages are widely adopted across data and software roles, making them essential for job seekers.
- Job seekers should prioritize learning TypeScript while considering Javascript and SQL to align with emerging employer demands.
- Job seekers should monitor evolving language trends to adapt their skill sets according to both current usage and emerging employer preferences.

DATABASE TRENDS





DATABASE TRENDS- FINDINGS & IMPLICATIONS

Findings

- Relational databases (e.g., MySQL, PostgreSQL) are the most commonly used databases in current job postings.
- There are minor differences between the databases used and the databases desired.
- The top databases used and desired largely overlap, but the ranking order differs slightly, indicating evolving preferences in the industry.

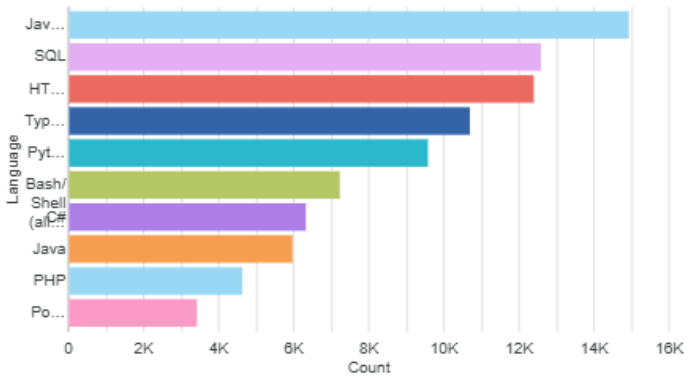
Implications

- Employers are requesting both relational databases and some cloud-based databases (e.g., MongoDB), indicating a demand that is somewhat broader than current usage trends.
- Proficiency in relational databases remains a core skill for job seekers
- Learning cloud or specialized database technologies can improve earning potential.

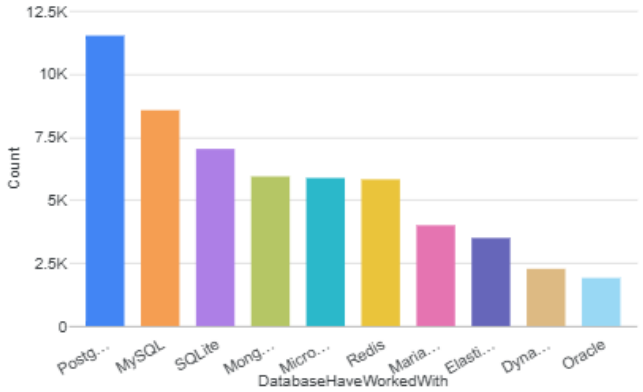
DASHBOARD TAB 1: Current Technology Usage

Current Technology Usage

Top 10 Languages Used

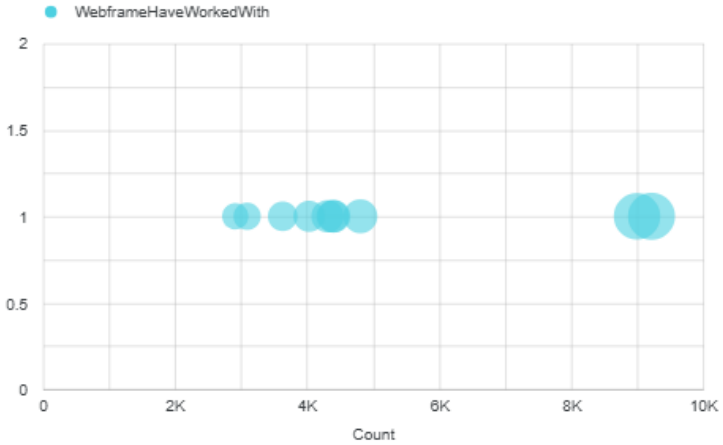


Top 10 Databases Used



Top 10 Platforms Used

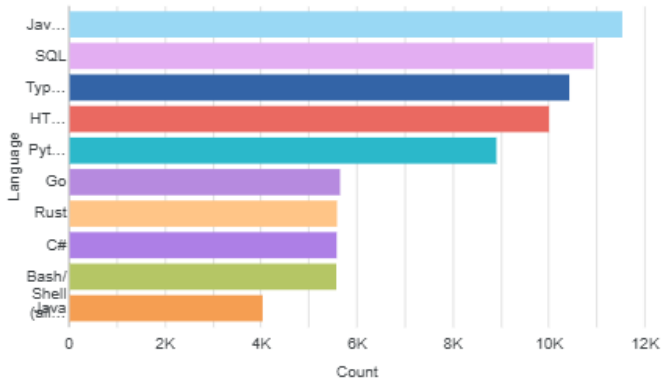
Top 10 Web Frameworks Used



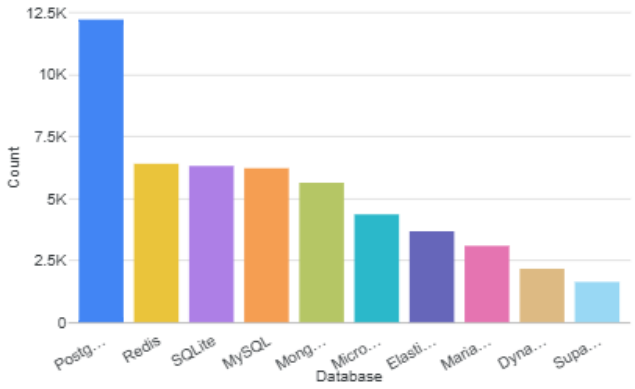
DASHBOARD TAB 2: Future Technology Trends

Future Technology Trends

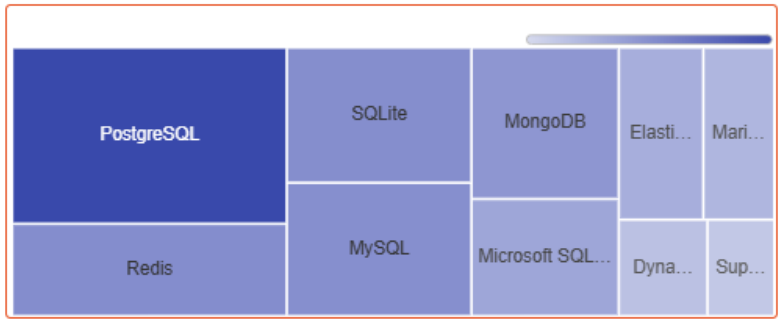
Top 10 Languages Desired Next Year



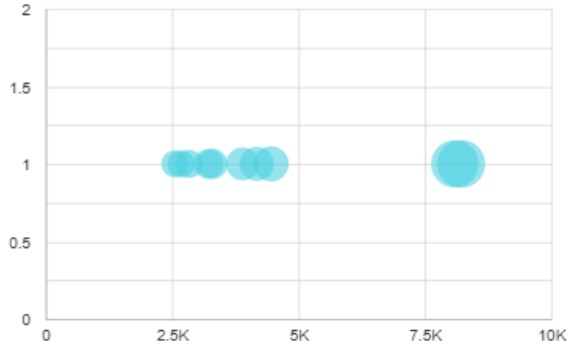
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



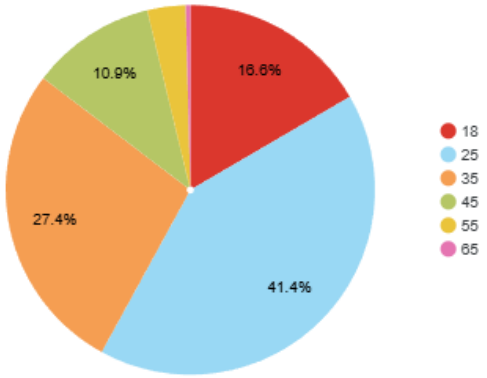
Top 10 Desired Web Frameworks



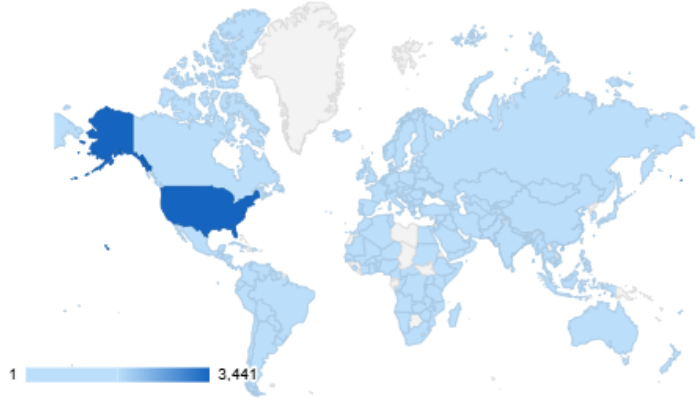
DASHBOARD TAB 3: Demographics

Demographics

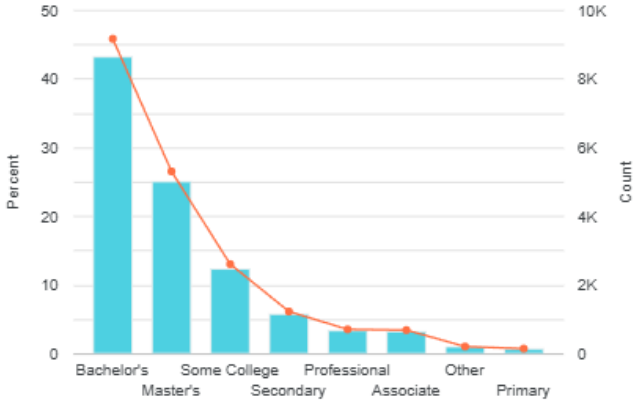
Respondents by Age



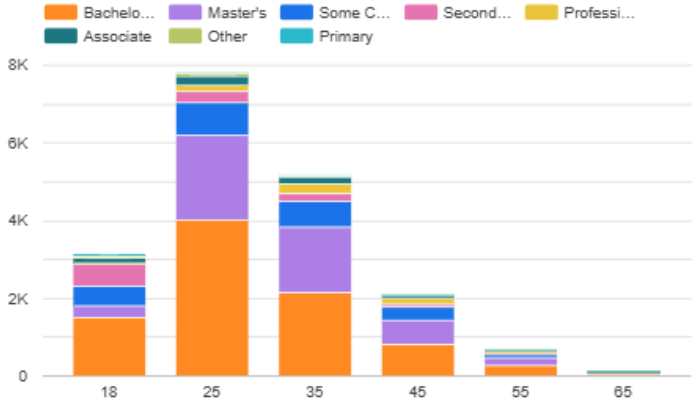
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



DISCUSSION

Programming Language Trends

- This shows a small difference between current usage and employer demand, and areas where job seekers can focus on developing their skills.

Database Trends

- Relational databases like MySQL and PostgreSQL dominate usage, but employer demand also includes NoSQL databases such as MongoDB. This highlights the importance of learning both traditional and emerging database technologies.

Job Skills Insights

- Employers prioritize candidates with a combination of foundational and specialized skills.

Salary Insights

- Average annual salary varies across programming languages and technical skills.High-demand and specialized languages are generally associated with higher salaries.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Certain programming languages (JavaScript, SQL, HTML/CSS) and databases (MySQL, PostgreSQL) dominate current usage.
- Employer demand emphasizes both foundational skills and emerging technologies (Python, SQL, MongoDB).
- Highly demanded and specialized skills are associated with higher salaries.

Implications

- These are essential foundational skills; job seekers should prioritize mastering them to be competitive.
- Job seekers may benefit from learning emerging technologies in addition to widely used ones to align with market needs.
- Focusing on these skills can increase earning potential and career opportunities.

CONCLUSION

•This project analyzed data from the Stack Overflow Developer Survey to uncover trends in programming languages, databases, job skills, and salary models.

•The findings highlight the importance of targeted skill development for job seekers. By focusing on technologies that are both widely used and in demand, individuals can increase their chances of finding employment and maximize career development opportunities.

•Overall, dashboards provide a comprehensive and interactive overview of the technology job market, enabling professionals to make informed decisions about skill acquisition and career planning.

APPENDIX

•Full Data Tables:

Top 10 Programming Languages & Databases Used and Desired, plus Average Annual Salary by Language

•Developer Survey Details:

Methodology, number of respondents, roles/fields covered